VIARTRIL®-S SACHET
Glucosamine Sulphate
Powder for oral solution

1. NAME OF THE MEDICINAL PRODUCT

Viartril-S Sachet

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Active substances: Crystalline glucosamine sulphate

Each sachet contains:

Crystalline glucosamine sulphate	1884 mg
equivalent to:	
Glucosamine sulphate	1500 mg
Sodium chloride	384 mg

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Single dose sachet.
Consists of white, crystalline and odourless powder for oral solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of the symptoms of osteoarthritis, i.e. pain and function limitation.

4.2 Posology and method of administration

The content of one sachet (dissolved in a glass of water) should be taken once daily, preferably at meals.

4.3 Contraindications

Hypersensitivity to glucosamine or to any of the excipients.

The product should not be given to patients who are allergic to shellfish as the active ingredient is obtained from shellfish.

4.4 Special warnings and precautions for use

In asthmatic patients, the product should be used with caution as these patients could be more susceptible to develop an allergic reaction to glucosamine with a possible exacerbation of their symptoms.

This medicinal product contains sodium. To be taken into consideration by patients on a controlled sodium diet.

No special studies were performed in patients with renal or hepatic insufficiency. The administration of the product to patients with severe hepatic or renal insufficiency should be under medical supervision.

Caution is advised in treatment of patients with impaired glucose tolerance. Closer monitoring of blood sugar levels may be necessary at the beginning of treatment.

Glucosamine should not be used in children and adolescents under the age of 18 since safety and efficacy have not been established.

The presence of other joint disease which would require alternative treatment should be excluded.

Excipients:

This medicinal product contains aspartame, a source of phenylalanine. It may be harmful if you have phenylketonuria (PKU), a rare genetic disorder in which phenylalanine builds up because the body cannot remove it properly.

This medicinal product contains sorbitol, a source of fructose. Patients with hereditary fructose intolerance (HFI) should not take/be given this medicine.

4.5 Interaction with other medicinal products and other forms of interaction

No specific drug interaction studies have been performed, however, the physico-chemical and pharmacokinetic properties of glucosamine sulphate suggest a low potential for interaction. In addition, glucosamine sulphate was found not to inhibit or to induce any of the major human CYP450 enzymes. In fact, the compound does not compete for absorption mechanisms and, after absorption, does not bind to plasma proteins, while its metabolic fate as an endogenous substance incorporated in proteoglycans or degraded independently of the cytochrome enzyme system, is unlikely to give rise to drug interactions.

An increased effect of coumarinic anticoagulants during concomitant treatment with glucosamine sulphate has been reported. Therefore, a closer monitoring of the coagulation parameters may be considered in these patients when initiating or ending glucosamine therapy.

The oral administration of glucosamine sulphate can enhance the gastrointestinal absorption of tetracyclines but the clinical relevance of this interaction is probably limited.

Steroidal or non-steroidal analgesic or anti-inflammatory agents can be administered together with glucosamine sulphate.

4.6 Fertility, pregnancy and lactation

Pregnancy
There is no adequate data from the use of glucosamine in pregnant women. Glucosamine should not be used during pregnancy.

Lactation
It is not known whether or not glucosamine is excreted in human milk. The use of glucosamine during lactation is not recommended as there is no data available on the safety of the newborn.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. No important effects on the CNS or motor system are known that might impair the ability to drive or use machines. However, caution is recommended if headache, somnolence tiredness, dizziness or visual disturbances are experienced.

4.8 Undesirable effects

The reported undesirable effects have been observed in a low proportion of patients. They are usually mild and transitory, which includes:

- ***Cardiovascular***: Peripheral oedema, tachycardia were reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.
- ***Central nervous system***: Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%).
- ***Gastrointestinal***: Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine.
- ***Skin***: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

Evaluation of the undesirable effects is based on the following frequency information:

Very common: $\geq 1/10$
Common: $\geq 1/100$ to $<1/10$
Uncommon: $\geq 1/1,000$ to $<1/100$
Rare: $\geq 1/10,000$ to $<1,100$
Very Rare: $<1/10,000$
Not known: cannot be estimated from the available data

System Organ Class	Common	Uncommon	Not Known
Immune system disorder			Allergic reaction (Hypersensitivity)
Metabolism and nutrition disorders			Diabetes inadequate control
Psychiatric disorders			Insomnia
Nervous system disorders	Headache Somnolence		Dizziness
Eye disorders			Visual disturbances
Cardiac disorders			Cardiac arrhythmias e.g. Tachycardia
Vascular disorders		Flushing	
Respiratory, thoracic and mediastinal			Asthma / Asthma aggravated
Gastrointestinal disorders	Diarrhea Constipation Nausea Flatulence Abdominal pain Dyspepsia		Vomiting

Hepatobiliary disorders			Jaundice
Skin and subcutaneous tissue disorders		Erythema Pruritus Rash	Angioedema Urticaria
General disorders and administration site conditions	Tiredness		Oedema / peripheral oedema
Investigations			Hepatic enzyme elevation Blood glucose increased Blood pressure increased INR flactuation

Cases of hypercholesterolemia have been reported, but a causal link has not been demonstrated.

4.9 Overdose

No cases of accidental or intentional overdose are known. Based on animal acute and chronic toxicological studies in animals, symptoms of toxicity are not likely to occur at doses up to 200 times the therapeutic dose.

If overdose occurs treatment should be discontinued, symptomatic and standard supportive measure should be adopted as required e.g. act to restrore the hydroelectrolytic balance.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: other anti-inflammatory and anti-rheumatic agents, non-steroidal anti-inflammatory drugs. ATC code: M01AX05

Mechanism of action:
Glucosamine sulphate is the sulphate salt of the endogenous amino-monosaccharide glucosamine, a normal constituent and preferred substrate for the synthesis of glycosaminoglycans and proteoglycans in cartilage matrix and synovial fluid.

Early *in vitro* studies have shown that glucosamine sulphate stimulates the synthesis of glycosaminoglycans and thus of articular cartilage proteoglycans. However, glucosamine sulphate has been more recently shown to inhibit the interleukin-1 β (IL-β) intracellular signaling pathway via blockade of the intracellular activation and nuclear translocation of nuclear factor kappa B (NF-kB) in the cartilage chondrocytes and other relevant cells.

Pharmacodynamic effects:
Early *in vitro* studies have demonstrated that glucosamine sulphate has anabolic and anti- catabolic effects on cartilage metabolism; sulphate ions may contribute to the pharmacological effects exerted by glucosamine by controlling the rate of glycosaminoglycan and proteoglycan synthesis and inhibiting cartilage degrading enzymes.

More recent studies have postulated that glucosamine sulphate decreases IL-1 β mediated effects, thus inhibiting a cascade of events that lead to joint inflammation and cartilage damage, such as the synthesis of metalloproteases, cyclooxygenase-2 and extracellular matrix proteins that are absent in normal cartilage, the release of nitric oxide and of prostaglandin E2, the inhibition of chondrocyte proliferation and the induction of cell death.

Differently from NSAIDS, glucosamine does not directly inhibit cyclooxygenase activities. Human chondrocyte cell models have shown that crystalline glucosamine sulphate inhibits IL-1 stimulated gene expression at glucosamine concentrations similar to or lower than those found in plasma and synovial fluid of knee osteoarthritis patients receiving the drug at the therapeutic dose of 1500mg once daily. Animal models confirmed the potential of glucosamine sulphate at human equivalent doses in delaying the progression of the disease and alleviating its symptoms.

Clinical efficacy and tolerability:
The safety and efficacy of glucosamine sulphate have been confirmed in clinical trials for treatment up to three years.

Short- and medium-term clinical studies have shown that the efficacy of glucosamine sulphate on osteoarthritis symptoms is evident already after 2-3 weeks from the beginning of administration. However, differently from NSAIDS, glucosamine sulphate has shown a duration of effect which ranges from 6 months to 3 years.

Clinical studies of daily continuous crystalline glucosamine sulphate treatment up to 3 years have shown a progressive improvement on the symptoms and a delay of the joint structural changes, as determined by pain radiography.

Glucosamine sulphate has demonstrated a good tolerability over both short-term and long-term treatment courses.

5.2 Pharmacokinetic properties

Absorption:
After oral administration of ^{14}C-labelled glucosamine, the radioactivity is rapidly and almost completely (about 90%) absorbed systemically in healthy volunteers. The absolute bioavailability of glucosamine in man after administration of oral glucosamine sulphate was 44%, due to first-pass effect of the liver. After oral administration of daily repeated doses of 1500 mg of glucosamine sulphate in healthy volunteers under fasting conditions, the maximum plasma concentrations at steady-state ($C_{max,ss}$) averaged 1602±426 ng/ml between 1.5-4 h (median: 3 h; t_{max}). At steady state, the AUC of the plasma concentration vs. time curve was 14564±4138 ng.h/ml. It is unknown if meals significantly affect the drug oral bioavailability. The pharmacokinetics of glucosamine are linear after once daily repeated administrations in the dose interval 750-1500 mg with deviation from linearity at the dose of 3000 mg due to lower bioavailability. No gender differenced were found in man with regard to the absorption and to the bioavailability of glucosamine. The pharmacokinetics of glucosamine was similar between healthy volunteers and patients with osteoarthritis of the knee.

Distribution:
Glucosamine is significantly distributed in extra-vascular compartments including synovial fluid with an apparent distribution volume 37-fold higher than the total body water in humans. Glucosamine does not bind to plasma proteins, so it is highly unlikely that glucosamine might produce displacement drug interaction when co-administered with other drugs that are highly bound to plasma proteins.

Metabolism:
The metabolite profile of glucosamine has not been studied because being an endogenous substance; it is used as a building block for the biosynthesis of articular cartilage compositions. Glucosamine is mainly

metabolized through the hexosamine pathway and independently of the cytochrome enzyme system. Crystalline glucosamine sulphate does not act as an inhibitor nor as am inducer of the human CYP450 isoenzymes including CYP 3A4, 1A2, 2E1, 2C9 and 2D6 even when tested at concentrations of glucosamine 300-fold higher than the peak plasma concentrations observed in man after therapeutic doses of crystalline glucosamine sulphate. No clinically relevant metabolic inhibition and/or induction interactions are expected between crystalline glucosamine sulphate and co-administered drugs that are substrates of the human CYP450 isoforms.

Excretion:
In man, the terminal elimination half-life of glucosamine from plasma has been estimated at 15h. After oral administration of ^{14}C-labelled glucosamine to humans, the urinary excretion of radioactivity was 10±9% of the administered dose while fecal excretion was 11.3±0.1%. The mean urinary excretion of unchanged glucosamine after oral administration in man was about 1% of the administered dose suggesting that the kidney and liver do not significantly contribute to the elimination glucosamine and/or of its metabolites and/or its degradation products.

Special population
Patients with renal or hepatic impairment
The pharmacokinetics of glucosamine were not investigated in patients with renal or hepatic insufficiency. Studies in renal impairment patient were considered irrelevant due to the limited contribution of the kidney to the elimination of glucosamine. Similarly, studies in subjects with hepatic impairment were not conducted given glucosamine metabolic fate as an endogenous substance. Therefore, for what described above and in light of the good safety and tolerability profile of glucosamine, no dose adjustment is considered necessary in subjects with renal or hepatic insufficiency.

Children and adolescents
The pharmacokinetics of glucosamine was not investigated in children and adolescents.

Elderly patients
No specific pharmacokinetic studies were performed in elderly however in the clinical efficacy and safety studies mainly elderly patients were included. Dose adjustment is not required.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Aspartame, Sorbitol, Macrogol 4000, Citric acid.

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please refer outer carton for expiry date.
The expiry date applies to the correctly stored product in the intact package.
Do not use the product after the expiry date indicated on the label.

6.4 Special precautions for storage

Do not store above 30°C.

Keep out of reach of children / Jauhi daripada kanak-kanak.

6.5 Nature and contents of container

Package of 4 and 30 sachets.

6.6 Special precautions for disposal

Not applicable.

7. MANUFACTURER

Rottapharm Ltd.
Damastown Industrial Park, Mulhuddart
Dublin 15, Ireland

8. PRODUCT REGISTRATION HOLDER

Mylan Healthcare Sdn Bhd
15-03 & 15-04, Level 15, Imazium,
No. 8, Jalan SS 21/37, Damansara Uptown,
47400 Petaling Jaya, Selangor, Malaysia.

9. DATE OF REVISION

2 May 2024